Why learning and teaching learning analytics is hard: An experience from a real-life LA course using LA methods

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Abstract. Learning analytics emerged more than a decade ago to harness the power of data to understand and optimize learning, learners' behavior, and learning environments. Ever since, the field has grown to encompass a diverse range of methods, research strands and traditions. Recent literature reviews tell us that most common applications of learning analytics include predictive analytics, social network analysis, sequence and process analysis, visualizations, and dashboards to mention a few. In the same vein, the research field has attracted several interdisciplinary researchers and practitioners from computer science, education, data science, engineering, administration, and from the education technology industry. Whereas such diverse backgrounds and perspectives bring a wealth of different perspectives to the field, it makes teaching and learning analytics hard to narrow down in a single course. This study reports on the analysis of students' approach to learning learning analytics, reflects on the insights that learning analytics offers, and makes recommendations for future researchers who are teaching or investigating similar courses.

Keywords: learning analytics, sequence analysis, data mining, psychological networks, computer science education

1 Introduction

Learning analytics (LA) may be defined as "the measurement, collection, analysis and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs" [1]. The field has been conceptualized to harness an opportunity where data, analytics methods and computational power have reached a level of abundance that was hard to ignore. Several rewarding examples from the industry have helped to kindle the surge in enthusiasm for learning analytics [2, 3]. Nevertheless, the field started with an imbalanced emphasis on data-driven analytics over theory-driven insights, which soon was rebalanced for more learning theory-driven research design, insights, and interpretation [4]. Since data is at the core of LA, data privacy, ethics, and security were embedded since the early discussions of the field [5]. As such, the field has grown larger, more complex, and more inclusive of different disciplinary themes and topics beyond data, and analysis, e.g., education, learning theories, education psychology, ethics, data

security, and decision-making. Such growth makes preparing future LA researchers or practitioners challenging on the students' and teachers' side [6, 7].
On the teacher's side, they need to have a large array of interdisciplinary expertise across several separate fields, e.g., computer science, and education. Such interdisciplinary expertise can hardly be found in a single teacher and therefore, teachers need to collaborate with other departments, universities or researchers [7]. Curricula has to be created collaboratively in the same way with different expertise to cover the intended course objectives [2]. On the students' side, they face the challenges of studying topics and themes that they are unprepared for, or have no previous knowledge of. For instance, technical students coming from computer science or engineering have to grapple with their lack of background knowledge on learning theories and pedagogy. For them, basic terms like learning strategies, engagement, motivation are not very clear. On the other hand, students coming from educational sciences have to go through technical terminology and methods that go beyond their limited quantitative training [7]. Given the expected difficulties, the interdisciplinary nature of the field, the recency of LA, it is important to understand how students regulate their language learning in online environments [8].

Whereas there is a deluge of papers that analyzed learning and teaching in courses of all kinds, there is hardly any study that has analyzed a LA course which is a gap we aim to bridge in this study. We use LA methods to analyze students' access to a blended course where the logs of their actions on the Learning Management System (LMS) to study students' approaches to LA to derive insights that could help learners, teachers, and course designers

2 Analysis of students' online learning

The analysis of students' online strategies has been an active area of inquiry since the initial days of the field of LA. Several methods have been implemented, e.g., unsupervised clustering of learning activities to group similar learning activities together, sequence analysis to study the students' temporal activities, process mining to study students' transitions and temporal pathways, and social network analysis to study students' interactions [2]. Each of the aforementioned offers a focused perspective on a certain side of students' behavior. Building on the previous insights and the accumulating body of literature, we use sequence and process analysis to analyze approaches to the LA course. Sequence analysis has a rich repertoire of techniques for statistical and visual analysis and integrates well with other statistical methods e.g., clustering. Process mining gives another perspective to students' transitions between activities in a more holistic way that augments the linear sequential aspect offered by sequence analysis [9].
The research questions of this study are therefore: What does the analysis of students' performance and activities tell us about their approach to LA and in particular, what are

the tactics students use? To what extent such tactics are associated with performance? What is the important tactic that drives students' engagement in LA?

3 Description of the course

The blended Learning Analytics course offered at the University of Eastern Finland provides a framework for understanding the field and how data has been used in education. The course addressed the principles of LA, and discussed the theoretical background behind LA and the concepts of big data, learning theories, and how they explain students' behavior. The course also covered the main steps and procedures of LA in detail, including data gathering, analysis, generation of insights, and reporting. The course discussed the main ethical and privacy issues related to LA, offered several examples of existing LA systems, and allowed students to get familiar with the authoritative literature in the field. The practical section of the course enabled students to practice basic methods of analysis of educational data using real-life examples and authentic datasets. These methods included visualization, social network analysis, sequence mining, process mining, and predictive analytics using accessible tools. The course was delivered face-to-face and the online section served as the platform for interactions in the group project, interaction with the teacher and peers, access for the course materials, zoom links, assignments and final project. By the end of the course, students are expected to have acquired the following learning objectives:

1. *Describe* the taxonomy of different terms used in the LA field, and *identify* the differences, usages, and applications of such terms.
2. *Recognize* the different theoretical models' underpinnings in the LA process, and *apply* such theories to different problems.
3. *Explain* different ethical and privacy challenges, *analyze* institutional policies, and *design* a policy as an example for the group project.
4. *Identify* the different steps in the analytics process, *explain* their steps, and use them to *design* the group project.
5. *Evaluate* LA examples regarding their design, ethics, and methodology, and offer an analysis and critique of their approach.
6. *Apply* the principles learned in the course to analyze several datasets, report on the findings and discuss the possible insights, as well as *evaluate* its usefulness and applicability in a real-life situation.
7. Design a proposal for a LA project – in collaboration with group members – that includes objectives, ethics, data collection, methods, analysis, and reporting.

The course had several assignments covering the course themes, methods and intended objectives. The literature review assignment required students to summarize literature findings in five selected articles to identify within the given article, e.g., data collection, methods, learning theories, reporting, and conclusions, as well as to offer a detailed critique of the ethical and theoretical grounding. Practical assignments

addressed the methods taught in the course, e.g., social network analysis, sequence analysis, process mining and predictive analytics. Each assignment was given a separate grade except for the literature review assignment, which had a separate grade for each sub-theme of the analysis, such as ethics, data, and learning theory sub-theme to test students' understanding. The course also had a final assignment that carried the highest weight (30% of the grade) and required students to combine, clean, and analyze three datasets, compute certain indices, develop informative visualizations, create a predictive model, and reflect on the analysis of the results and how they can be applied in practice. The collaborative module (referred to as group work) was evaluated based on students' involvement in the discussion, the final project results as well as by peers of all other groups.

4 Methods

Students' Moodle logs were cleaned from non-learning activities (e.g., profile clicks) and teacher and administrator clicks (grading events, lecture uploads, etc.). The time-ordered logs of students' activities were re-coded to balance granularity with meaningfulness, i.e., to group logs that essentially signify the same action. As such, activities within the collaborative project were all coded as *collaboration*, log activities related to feedback were coded as *feedback*, logs of students' access to practical resources or assignments were coded as *practicals*, social interactions that are unrelated to learning were coded as social etc. The re-coded logs were grouped into sessions (groups of actions that happen without considerable interruption) and a sequence object was created. Sequence analysis was performed using the TraMineR package [10]. A sequence object was created for each session, plotted and clustered using the methods described in Saqr et al. [11] to identify students' tactics (typical sequences of interactions with the LMS). A regularized partial correlation network was estimated to estimate the association of the frequency of sessions with the final grade according to the methods of [12]. In partial correlation networks, nodes are correlated after conditioning for all other variables in the network similar to regression analysis. The expected influence centrality measures were also calculated to estimate which of the tactics stimulated engagement with other tactics and or students' achievement.

5 Results

Throughout the course, students demonstrated diverse levels of engagement in various activities (Table 1). The average number of clicks on assignments was 35.98, indicating active participation, with scores ranging widely (SD = 24.66). Collaborative work stood out, as students displayed impressive engagement with group work with a mean count of 268.69 clicks per student (SD = 30). Seeking and incorporating feedback was also evident, with a mean count of 26.93 (SD = 16.52). Students engaged with assignment instructions to a similar extent than with the assignments themselves, as shown by a

mean count of 36.13 (SD = 35.12), while self-monitoring efforts were notable with a mean count of 91.62 (SD = 16.52). The practical aspects of the course saw significant involvement, as students achieved a mean count of 81.96 (SD = 36.12). Utilization of the learning resources was consistent, as indicated by a mean count of 32.96 (SD = 31.18), while social interactions related to the course averaged at 13.24 (SD = 13.98). Viewing the course main page remained a focal point, with an average count of 210.02 (SD = 159.82).

Table 1. Descriptive statistics of the number of actions per student

Action	M	SD	MED
Assignment	35.98	24.66	30
Collaboration	268.69	175.71	230
Feedback	26.93	16.52	23
Instructions	36.13	35.12	27
Monitoring	91.62	44.30	88
Practicals	81.96	36.12	75
Resources	32.96	31.18	25
Social	13.24	13.98	9
View course	210.02	159.82	177

The distribution of the assignment grades (Fig. 1) showed some interesting results. Grades in learning theory were the lowest and more skewed toward lower grades. Learning analytics basics and ethics were relatively lower and distributed across a wide range of grades. On the other hand, group collaboration, and practical grades were higher and skewed to higher grades. Put another way, students were more interested in the practical and technical parts of the course, and less interested in theoretical parts, e.g., learning theory, basics, and ethics.

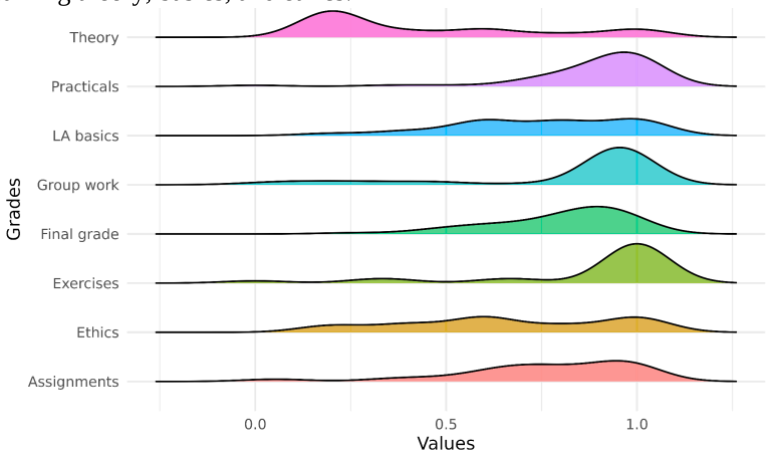


Fig. 1. Density plots of each grade (grades were normalized between 0 and 1 for comparison)

Using sequence clustering, we identified five clusters that are typical of students' online activities (**Fig. 2**). The first cluster (337 sessions, 11%) was labeled as *practicing* as it was dominated by actions involving practical activities. The second cluster was labeled *diverse* (1039 sessions, 34 %), as it was not dominated by any particular action and represents mixed episodes of multiple actions in the same session. A third cluster (913 sessions, 30%) was labeled *groupwork* as it was dominated by interactions in the forums doing group work. The fourth cluster (119 sessions, 4%) was dominated by feedback actions and, thus, dubbed *feedback*. The fifth cluster (668 sessions, 22 %) was dubbed *engagement and monitoring*, as it was dominated by access to learning resources, assignments, and monitoring activities.

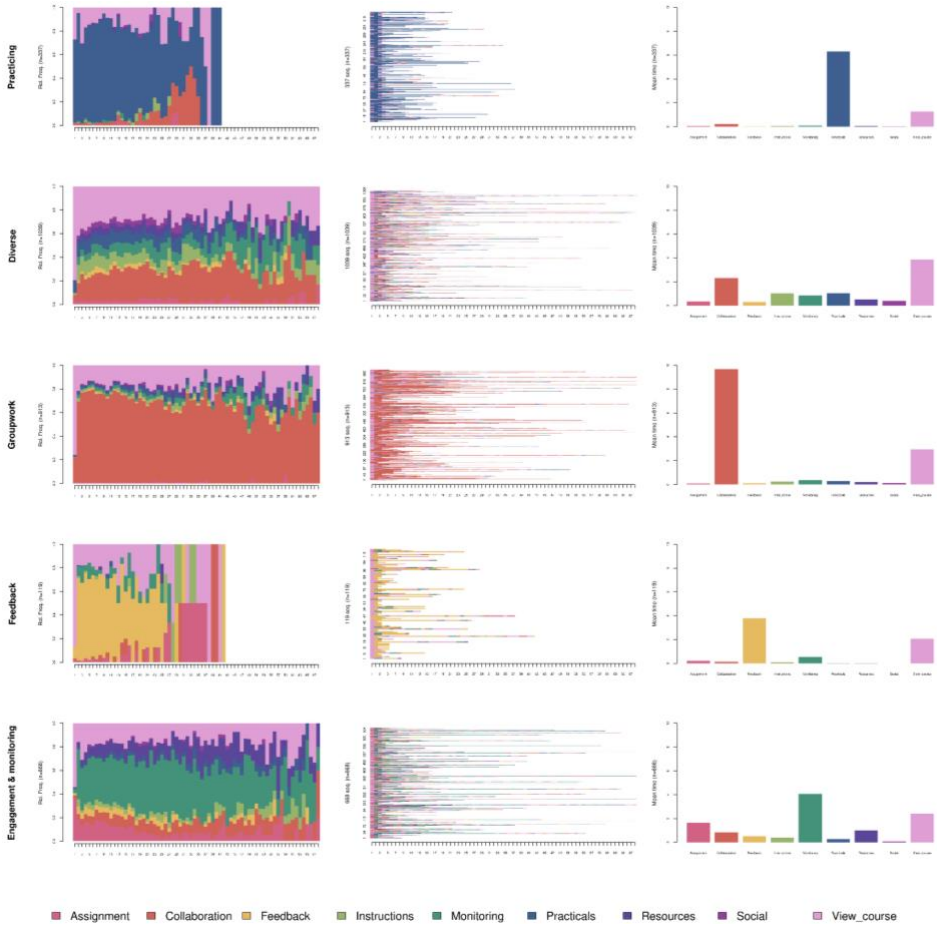


Fig. 2. Sequence distribution plot (left), sequence index plot (middle), mean time plot (right) of each of the five identified clusters

To identify which tactic is associated with better performance, we opted for a regularized partial correlation where a correlation is reported only and if the correlation is true after conditioning for all other variables similar to regression. The results in **Fig. 3** show that collaborative group work (*groupwork* tactic), *feedback* tactic and *diverse*

tactics were the most correlated with final grade. *Engagement & monitoring* tactics were highly correlated with *diverse* tactics and with *practicing* tactics but neither were correlated with grades after controlling for all other tactics. In other words, students who use feedback, learn with others and consume course activities are likely to score higher.

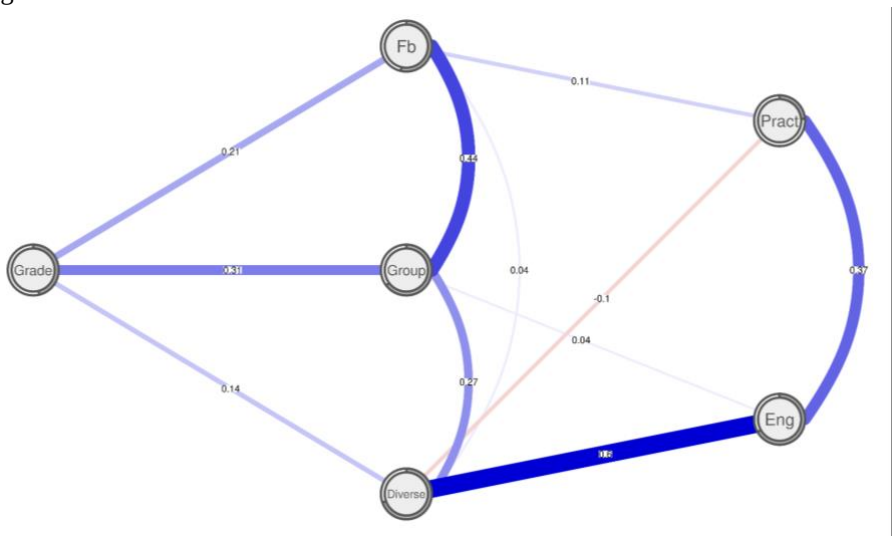


Fig. 3. Psychological network of the association between session types and the final grade (Fb = Feedback, Group = Group work, Eng = Engagement and monitoring, Pract = Practicing)

The centrality measures (**Fig. 4**) show that the *groupwork* tactic was the most central tactic that has driven all other activities positively in the network, followed by *engagement & monitoring* and *diverse* tactics. In turn, *practicing* sessions (since they were required from all) had a lower centrality compared to others activities as it only influenced the engagement tactic.



Fig. 4. Expected influence centrality of each tactic type and final grade

5 Discussion

Whereas the field of LA has grown rapidly as a research field, it has grown relatively slower in practice. Several factors have contributed to the slower adoption of LA that includes technical problems, data issues, legal and ethical challenges and shortage of interdisciplinary expertise [13, 14]. Therefore, training for future LA expertise has emerged in the form of traditional university courses, workshops and massive open online courses to mention a few. Yet, little is known about this relatively new discipline and how students' approach their learning, which is a gap this study aimed to bridge.

Our study revealed that technical students learning LA face challenges in the theoretical and educational sections of the course (ethics, learning theories and LA literature). The grade distribution showed clearly that these areas are widely distributed with obvious skewness towards lower grades. On the contrary, students showed higher interest in the technical and practical sections of the course. The grades of these sections showed clearly higher grades and less variance skewed to higher grades of these technically oriented sections.

Collaborative group work was the most important learning resource in our course, it has driven engagement more than any other activity (as per the expected influence centrality measure), correlated with the grades above and beyond all other activities and has been the sequence pattern with most number of activities. This may signify that in courses where students may have difficulty with some course content, collaboration and exchanging perspectives can facilitate, support and embolden students to learn. Whereas diverse sessions (content sessions) correlated with performance, we see that feedback has been instrumental. Feedback sessions, although few, contributed to higher grades (conditioned on all other variables including content); which is an indication that the time spent in feedback is very valuable and fruit bearing.

Learning analytics as a teaching endeavor is far from easy and has its challenges. As the field grows larger and more diverse, the challenges are not fading but are expected to grow. We have —by doing this analysis— shed a light on some small aspects of learning an interdisciplinary analytics course. We hope that future examples emerge, telling different aspects of the story. Only then we can build a body of knowledge that enables us to design better future courses. Nonetheless, our course study has some implications. First, it is important to prepare students for content they are not used to, e.g., learning, and educational theory. Second, collaborative work can help students overcome challenging course materials. Third, feedback is a central component of successful learning. Fourth, methods that emphasize the depth of learning rather than "assessment and assignment" driven learning should be implemented, e.g., collaborative learning, interactive and project-based learning as well as critical reflection.

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